# AURA HEALTH (HealthTrack+ for Women)

Project PPT Audit, Critical Corrections & Doctor Portal Presentation Guide

AI-Powered Menstrual Health, Rotterdam v4 Diagnostics & Web Doctor Portal Ecosystem

## 1. Executive Overview

This document provides an actionable, slide-by-slide modification roadmap for your seminar/final presentation. Aligning your PPT slides with the actual codebase eliminates red flags during examiner viva questions, while highlighting your strongest technical innovations: the Web Doctor Portal and On-Device Rotterdam Edge ML.

## 2. Top 4 Critical Red Flags (Must Fix to Prevent Viva Discrepancies)

### A. Frontend Framework (Slide 20, 21): React Native / Expo -> React 19 + Capacitor

PPT states "React Native (Expo)". The codebase is built with React 19, TypeScript, Vite, Tailwind CSS, and Capacitor (with native Android Kotlin plugin). If examiners look at your package.json, there is no Expo runtime. Change all references to "React 19 with Vite & Capacitor Native Android".

### B. Database Engine (Slide 20): MySQL -> Google Cloud Firestore + Local Cache

PPT lists "MySQL, Firebase". Your project exclusively uses Google Cloud Firestore (NoSQL) with offline caching and an Abstract Repository Pattern (Firestore / LocalStorage). Mentioning MySQL invites questions about SQL tables and schemas that do not exist.

### C. ML Inference (Slide 8, 9, 20): Cloud Python API -> On-Device Edge ONNX Model

PPT suggests a remote Python FastAPI server. In reality, you have implemented on-device ONNX Runtime (pcos_rotterdam_v4.onnx) running directly on Android via OnnxPredictorPlugin.java. On-device inference guarantees zero network latency and 100% patient data privacy. Emphasize this!

### D. Project Branding Consistency: HealthTrack+ -> Aura Health (HealthTrack+)

The Android package is com.aura.health and all UI assets use "Aura Health". Unify the title and intro slides to: "Aura Health (HealthTrack+ for Women)" so both project identifiers match harmoniously.

## 3. Doctor Portal Website: How to Represent It in the PPT

Your project features a dual-frontend ecosystem: the Patient Mobile App and the Doctor Web Portal. In your presentation, highlight these 5 c

- **1. Verified Clinician Authentication:** Role-Based Access Control (RBAC) ensuring only credentialed physicians can review patient medical records.
- **2. Patient Health Record Stream:** Real-time visibility into consented patient cycle histories, flow intensity, and basal body temperature biphasic curves.
- **3. Rotterdam Phenotype Diagnostic Viewer:** Displays AI-classified Rotterdam Phenotype (A, B, C, or D), model confidence percentage, and 19 biomarker attributions (WHR, mFG score, androgenic signs).
- **4. Standardized Clinical PDF Report Ingestion:** Seamless ingestion and review of the multi-cycle clinical summary PDF generated by the patient application.
- **5. Digital Consultation Notes & Lifestyle Rx:** Enables clinicians to issue dietary plans, seed cycling guidance, and medical follow-up advice directly to the patient dashboard.

## 4. Slide-by-Slide Exact Action Items

**Slide 3: Introduction**

Change text to emphasize the dual-portal ecosystem: "Aura Health integrates on-device menstrual cycle tracking, edge AI Rotterdam PCOS screening, and a dedicated Web Doctor Portal for physician collaboration."

**Slide 5: Research Gap**

Emphasize that existing apps lack a direct clinician bridge. Your Doctor Portal closes this gap by allowing patients to share multi-cycle diagnostic reports with their doctors.

**Slide 8: Methodology**

Update Step 2 to mention "Multi-Role RBAC (Patients & Doctors) via Firebase Auth", and Step 4 to "On-Device ONNX Machine Learning Inference with 19 clinical biomarkers".

**Slide 9: Proposed System Architecture**
In the Mobile box: indicate "ONNX Runtime (On-Device Inference)". In the Doctor Portal box: list "Patient Records, Phenotype Analysis, Prescriptions". In Database: replace MySQL with "Cloud Firestore".

**Slide 10: Dataset & Feature Schema**
Add the 19 Rotterdam features: Age, Menarche Age, BMI, Waist-to-Hip Ratio (WHR), Mean Cycle, Variance, mFG Hirsutism (0-36), Acne, Alopecia, Acanthosis, Birth Control override, and 4 Lab parameters.

**Slide 13: SRS Requirements**
Under Functional: add "Doctor Portal Access & Consultation Notes". Under Non-Functional: emphasize "Edge ML Privacy" and "Offline-First LocalStorage Fallback".

**Slide 14-17: Use Case & DFDs**
Ensure the Doctor actor connects to: "Log in to Doctor Portal", "View Patient Profile & Rotterdam Diagnosis", "Issue Digital Prescription / Notes", and "Manage Appointments".

**Slide 20: Frameworks & Tech Stack**
Replace React Native/Expo with: React 19, TypeScript, Vite, Tailwind CSS, Capacitor (Android Native Kotlin Plugin). Replace MySQL with: Google Cloud Firestore (NoSQL).

**Slide 21: Design Details**
Detail the Decoupled Client-Server & Edge Hybrid Architecture: On-device ONNX inference for privacy, Cloud Firestore for sync, and Web Dashboard for doctors.

## 5. Anticipated Viva / Examiner Questions & Winning Answers

**Q1: Why did you build both a Mobile App and a Doctor Web Portal?**
Patients need a fast, private mobile app for daily symptom and cycle logging on the go. Doctors need a desktop-friendly widescreen dashboard to review multi-cycle longitudinal charts, Rotterdam phenotype risk scores, and clinical PDF reports during medical consultations.

**Q2: Why use ONNX Runtime on the phone instead of calling a Python server?**
Patient menstrual and hormonal data is highly sensitive (HIPAA/GDPR privacy). Transmitting raw symptoms to a remote server creates privacy vulnerabilities and requires internet. Our custom Kotlin plugin executes pcos_rotterdam_v4.onnx on-device in under 30ms, completely offline.

**Q3: How does your PCOS diagnosis differ from standard cycle trackers?**
Standard trackers only look at missed days. Aura Health implements the international 2003/2018 Rotterdam Consensus Guidelines, evaluating 19 parameters including hyperandrogenism (mFG hirsutism, cystic acne, alopecia) and categorizing patients into Phenotypes A, B, C, or D.